

Refutation of Conjecture 00000000277: the maximum Laplace multiplicity on a flat torus is not 6

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Abstract

We disprove Conjecture 00000000277. The conjecture asserts that M , the maximum multiplicity of a Laplace eigenvalue on a planar flat torus (supremum over all such tori), equals 6, and that multiplicity six is attained only on a lattice torus whose automorphism group has order thirty-two. Under the literal reading of M (maximum over *all* eigenvalues) the claim fails outright: on the square torus $\mathbb{R}^2/\mathbb{Z}^2$ the eigenvalue $4\pi^2 N$ has multiplicity $r_2(N)$, the number of representations of N as a sum of two squares, and $r_2(5^k) = 4(k+1) \rightarrow \infty$, so the multiplicity is unbounded and $M = \infty$. Independently, the order-thirty-two clause is impossible under *either* reading: the crystallographic restriction forces every lattice-preserving rotation to have order in $\{1, 2, 3, 4, 6\}$, so the automorphism group of a planar lattice has order at most 12. We are careful to record that if M had been intended as the multiplicity of the *first non-zero* eigenvalue only, then the value 6 is correct (attained by the hexagonal lattice); nevertheless the statement as written is false.

1 The conjecture

Conjecture 1 (00000000277). $M(T)$ is the maximum multiplicity of a Laplace eigenvalue on a planar flat torus $T = \mathbb{C}/\Lambda$, and M is its supremum over all flat tori. Then $M = 6$ (six is known to be attainable, and upper bounds ≥ 6 are constrained by congruence structure); moreover multiplicity six is attained only on the lattice torus with maximal automorphism group of order thirty-two.

We show that this statement is false. Section 2 sets up the spectrum of a flat torus, Section 3 produces unbounded multiplicities on the square torus, Section 4 rules out the order-thirty-two clause, and Section 5 states precisely the sense in which the number 6 is nevertheless correct.

2 The spectrum of a flat torus

Let $\Lambda \subset \mathbb{R}^2 \cong \mathbb{C}$ be a lattice of full rank and let $T = \mathbb{R}^2/\Lambda$ be the associated flat torus, equipped with the flat metric induced by the Euclidean metric on $\mathbb{R}^2$. The Laplace–Beltrami operator $-\Delta_T$ has spectrum

$$\{4\pi^2|k|^2 : k \in \Lambda^*\}, \quad \Lambda^* = \{k \in \mathbb{R}^2 : \langle k, \lambda \rangle \in \mathbb{Z} \text{ for all } \lambda \in \Lambda\}, \quad (1)$$

where Λ^* is the dual lattice, and the multiplicity of the eigenvalue $4\pi^2|k|^2$ is the number of dual-lattice vectors of that squared length:

$$\text{mult}_T(4\pi^2 N) = \#\{k \in \Lambda^* : |k|^2 = N\}. \quad (2)$$

This is the standard Fourier expansion: the functions $x \mapsto e^{2\pi i \langle k, x \rangle}$ for $k \in \Lambda^*$ form a complete orthogonal system of eigenfunctions with eigenvalues $4\pi^2 |k|^2$. In particular each multiplicity is finite and even whenever the length is non-zero, because k and $-k$ are distinct dual vectors of equal length.

For the square torus $\Lambda = \mathbb{Z}^2 = \mathbb{Z}[i]$, the lattice is self-dual, $\Lambda^* = \mathbb{Z}^2$, and the squared lengths are exactly the integers representable as a sum of two squares:

$$\text{mult}_{\mathbb{Z}[i]}(4\pi^2 N) = \#\{ (a, b) \in \mathbb{Z}^2 : a^2 + b^2 = N \} =: r_2(N). \quad (3)$$

3 Unbounded multiplicity on the square torus

The arithmetic function $r_2(N)$ is classical. Since $\mathbb{Z}[i]$ is a unique factorisation domain, the number of representations of N as a sum of two squares is

$$r_2(N) = 4(d_1(N) - d_3(N)), \quad (4)$$

where $d_1(N)$ (resp. $d_3(N)$) counts the positive divisors of N congruent to 1 (resp. 3) modulo 4; this is Jacobi's two-square theorem. We will only need the case $N = 5^k$, whose divisors are $1, 5, 5^2, \dots, 5^k$, all congruent to 1 mod 4. Hence

$$r_2(5^k) = 4(k+1) \xrightarrow[k \rightarrow \infty]{} \infty. \quad (5)$$

The first values are

$$r_2(1) = 4, \quad r_2(5) = 8, \quad r_2(25) = 12, \quad r_2(125) = 16, \quad r_2(625) = 20, \quad r_2(3125) = 24. \quad (6)$$

Concretely $r_2(25) = 12$ comes from $(a, b) \in \{(\pm 5, 0), (0, \pm 5), (\pm 3, \pm 4), (\pm 4, \pm 3)\}$, and in general $r_2(5^k) = 4(k+1)$ counts the pairs $(\pm 2^i 5^j, \pm 2^{i'} 5^{j'})$ with $i + i' = j + j' = k$.

Proposition 2 (Unbounded multiplicity; $M = \infty$). *For every m there is a planar flat torus carrying a Laplace eigenvalue of multiplicity at least m . Consequently the supremum M of all such multiplicities is $+\infty$, and in particular $M \neq 6$.*

Proof. Take the square torus $T = \mathbb{R}^2/\mathbb{Z}^2$ and $N = 5^k$ for $k \geq \lceil m/4 \rceil - 1$. By (3) the eigenvalue $4\pi^2 N$ has multiplicity $r_2(5^k) = 4(k+1) \geq m$. Thus multiplicities are unbounded, so the supremum is infinite. Already for $k = 2$, i.e. $N = 25$, the multiplicity is $r_2(25) = 12 > 6$, so the claimed value $M = 6$ fails on a single eigenvalue. $\square$

Remark 3. Proposition 2 addresses the literal reading of the definition, in which $M(T)$ is the maximum multiplicity over all eigenvalues of $-\Delta_T$. The conclusion $M = \infty$ is not a consequence of some exotic lattice: the square torus, the simplest and most classical flat torus, already witnesses multiplicities $12, 16, 20, 24, \dots$.

4 The order-thirty-two lattice is impossible

The second clause of the conjecture asserts that multiplicity six is attained only on a lattice torus whose automorphism group has order thirty-two. This clause is false under either reading of M , for a reason independent of the spectral computation above.

Let $\Lambda \subset \mathbb{R}^2$ be a lattice and let $\text{Aut}(\Lambda) = \{g \in O(2) : g\Lambda = \Lambda\}$ be its orthogonal automorphism group, a finite group. Every $g \in \text{Aut}(\Lambda)$ is a rotation or a reflection; we consider the rotations.

Theorem 4 (Crystallographic restriction). *If $g \in O(2)$ has finite order n and preserves a full-rank lattice $\Lambda \subset \mathbb{R}^2$, then the trace $\text{tr}(g) = 2 \cos(2\pi/n)$ is an integer, and therefore*

$$n \in \{1, 2, 3, 4, 6\}. \quad (7)$$

Proof. In an integral basis of Λ , the map g is represented by a matrix $A \in GL_2(\mathbb{Z})$. Since g has order n , so does A , hence $A^n = I$. The trace $\text{tr}(A)$ is an integer, and it equals the sum of the two eigenvalues of A over $\mathbb{C}$. Those eigenvalues are roots of unity ζ, ζ^{-1} with $\zeta^n = 1$, so for a rotation of order n , $\text{tr}(A) = \zeta + \zeta^{-1} = 2 \cos(2\pi/n)$ up to the choice of the primitive root. Thus $2 \cos(2\pi/n) \in \mathbb{Z}$. Since $-2 \leq 2 \cos(2\pi/n) \leq 2$, the only possibilities are $2 \cos(2\pi/n) \in \{-2, -1, 0, 1, 2\}$, which occur exactly at $n \in \{1, 2, 3, 4, 6\}$. $\square$

Corollary 5. *No planar lattice has an automorphism of order thirty-two; the largest possible order of $\text{Aut}(\Lambda)$ is 12, attained by the hexagonal lattice (whose automorphism group is the dihedral group of order 12). In particular, there is no “order-thirty-two lattice torus”, so the second clause of Conjecture 00000000277 is false.*

Proof. By Theorem 4 the rotational part of $\text{Aut}(\Lambda)$ is contained in the rotations of order dividing 6, and reflections are involutions. A finite subgroup of $O(2)$ generated by these is either cyclic of order at most 6 or dihedral of order at most $2 \cdot 6 = 12$. The bound 12 is achieved by the hexagonal lattice, whose automorphism group is the dihedral group D_6 of order 12. In particular 32 is not the order of the automorphism group of any planar lattice. $\square$

5 The honest qualification: the first-eigenvalue reading

We must distinguish two readings of M .

1. *Literal reading (maximum over all eigenvalues).* As in Conjecture 00000000277, $M(T)$ is the maximum multiplicity of a Laplace eigenvalue, i.e. over all of $\text{spec}(-\Delta_T)$. Under this reading Proposition 2 gives $M = \infty \neq 6$: the claim is false, and moreover no finite value can be correct.
2. *First-eigenvalue reading (multiplicity of λ_1 only).* If instead M were defined as the multiplicity of the *first non-zero* eigenvalue λ_1 , then the value 6 is correct: it is attained by the hexagonal (triangular) lattice, and the geometric argument (all shortest dual vectors have equal length, pairwise angles at least 60° , and at most six points on a circle can be 60° -separated) shows that no flat torus has λ_1 -multiplicity exceeding 6. This is the classical statement that the two-dimensional kissing number is 6.

The conjecture is nevertheless false as written, for two separate reasons.

- Under the literal reading, $M = \infty$ (Proposition 2), so the asserted value $M = 6$ is wrong.
- Under *both* readings, the clause “multiplicity six is attained only on the order-thirty-two lattice torus” is impossible: by Corollary 5 no planar lattice has automorphism group of order 32. Even in the first-eigenvalue reading, where six is attainable, it is attained by the hexagonal lattice with automorphism group of order 12, not by any order-thirty-two lattice.

Thus Conjecture 00000000277 is false as stated.

References

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